

LECTURE 5

SOCRATES END CLASS OF ANATOMY

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KNEE JOINT

The knee is a **compound synovial joint**, meaning more than two bones and more than one articulation are involved.

It articulates:

- **Superiorly:** with the **medial and lateral femoral condyles**
- **Inferiorly:** with the **medial and lateral tibial condyles**, **WITH MENISCI INTERPOSED**
- **Anteriorly:** with the **patella**, which articulates with the femur

So functionally, this is **one joint capsule**, but **two different joint mechanics** inside it.



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TYPES OF JOINTS INSIDE THE KNEE

There are **two articulations**:

1) TIBIOFEMORAL JOINT (FEMUR ↔ TIBIA)

- **Modified hinge joint**¹
- Allows:
 - Flexion & extension
 - A small degree of rotation (important for locking/unlocking)

Calling it a “**pure hinge**²” is a lie the examiners want you to believe — rotation is essential here.

2) PATELLOFEMORAL JOINT (PATELLA ↔ FEMUR)

- **Plane (gliding) joint**
- Patella slides up and down the femoral trochlea during knee movement
- Increases the **lever arm of quadriceps**, improving extension power

¹ A modified hinge joint primarily allows flexion and extension but is structured in a way that permits a limited degree of movement in other directions, such as slight rotation or side-to-side motion, when the joint is in a specific position (usually flexed).

² A pure hinge joint, or true hinge joint, functions like a mechanical door hinge, with movement restricted to a single axis or plane (uniaxial). The articulating bone surfaces fit together perfectly to ensure this limitation

WHY DOES THE KNEE NEED SO MANY LIGAMENTS?

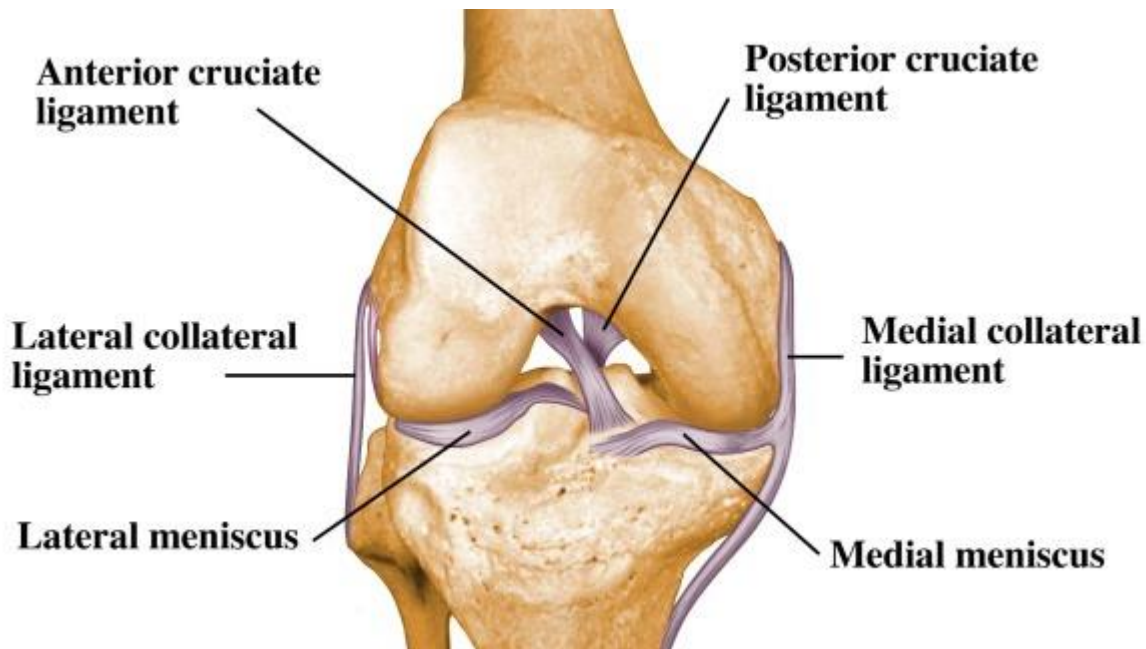
Because the **articular surfaces are incongruent**(Not perfectly fit):

- Femoral condyles are large and rounded
- Tibial condyles are relatively flat

Without ligaments and menisci, the femur would skateboard off the tibia.

So stability comes mainly from:

- **Ligaments**
- **Menisci**
- **Muscles (especially quadriceps & popliteus)**



INTRACAPSULAR STRUCTURES (INSIDE THE FIBROUS CAPSULE)

CRUCIATE LIGAMENTS (THE “X” INSIDE THE KNEE)

They are called **cruciate** because they **cross each other**.

ANTERIOR CRUCIATE LIGAMENT (ACL)



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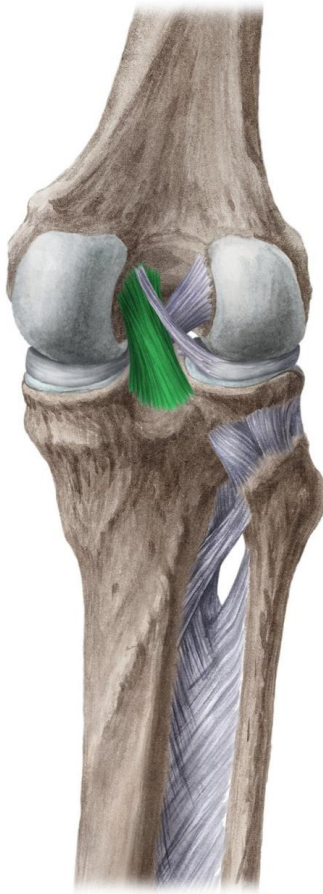
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- **Tibial attachment(O):** anterior intercondylar area of tibia
- **Femoral attachment(I):** medial surface of the **lateral femoral condyle**
- **Direction:** from **anterior tibia** → **upward, backward, lateral**(Like in the above pic)

Function (high-yield wording):

- Prevents anterior displacement of Tibia on femur (or posterior displacement of femur on tibia)
- Prevents hyperextension
- Major stabilizer during sudden stops & direction changes
- **Trick by Beka:** Just relate the ACL with Tibia (prevents Anterior displacement of Tibia...) and also for PCL (prevents posterior displacement of Tibia...)
- So, if it's femur, it will be inversed!

POSTERIOR CRUCIATE LIGAMENT (PCL)



- **Tibial attachment(O):** posterior intercondylar area of tibia
- **Femoral attachment(I):** lateral surface of the **medial femoral condyle**
- **Direction:** **from posterior tibia → upward, forward, medial**

Function:

- Prevents **posterior displacement of tibia** on femur (or anterior displacement of femur on tibia)
- Strongest ligament of the knee

Clinical pearl:

When **walking downstairs**, gravity pushes the femur **anteriorly** → **PCL resists this.**

TRANSVERSE LIGAMENT

- Connects **anterior horns of medial & lateral menisci**
- Helps coordinate their movement during flexion/extension



MENISCI (FIBROCARILAGE, NOT LIGAMENTS)

There are **two menisci**, sitting on the tibial condyles.

MEDIAL MENISCUS



- **C-shaped**
- Firmly attached to:
 - Joint capsule
 - **Deep fibers of the Medial Collateral Ligament (MCL)**
- Less mobile → **more frequently injured**

LATERAL MENISCUS



- **Almost circular (O-shaped)**
- **More mobile**
- **NOT attached to LCL**(Lateral collateral lig) → this is crucial
- Safer during rotational injuries

FUNCTIONS OF MENISCI (THINK “S-L-I-D”)

- **Shock absorption**
- **Load distribution**
- **Increases joint congruency**(So Fit in perfect)
- **Distributes synovial fluid**

EXTRACAPSULAR LIGAMENTS (OUTSIDE THE CAPSULE)

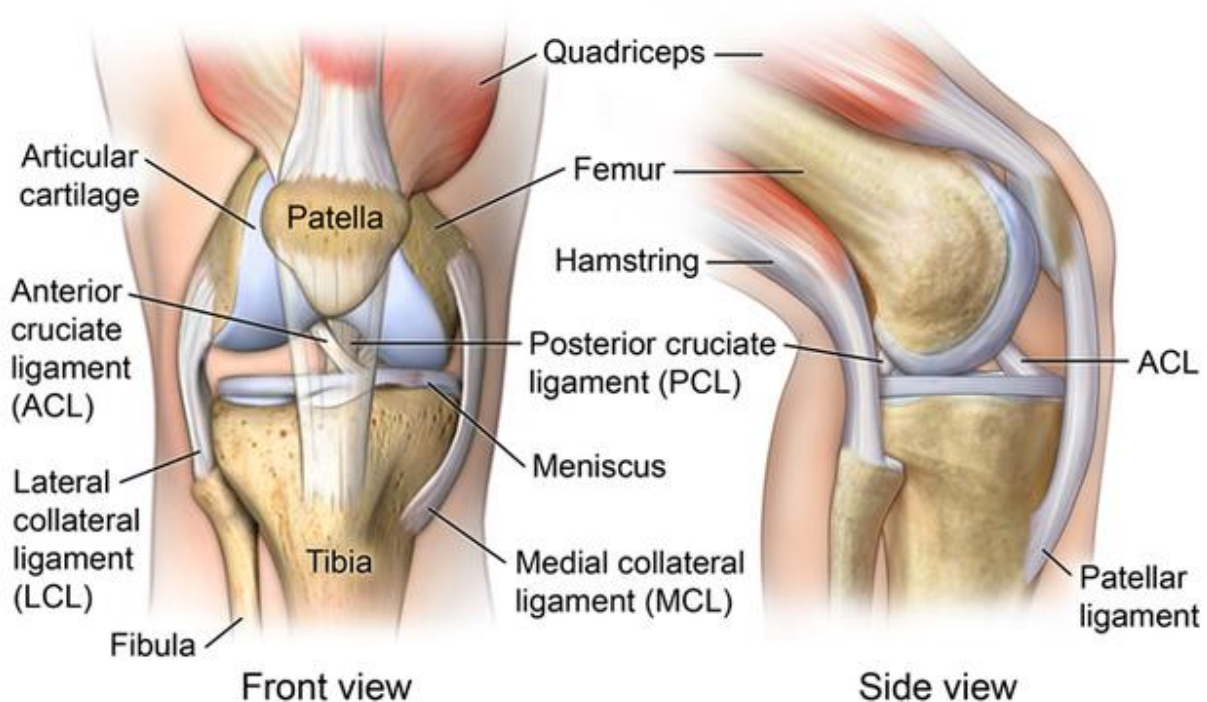
MEDIAL COLLATERAL LIGAMENT (MCL / TIBIAL COLLATERAL)

- Femur → tibia
- Attached to **medial meniscus**
- Injuries often involve **MCL + medial meniscus**

FIBULAR COLLATERAL LIGAMENT (LCL)

- Femur → fibula
- **Not attached to lateral meniscus**
- Can be examined clinically by varus stress test

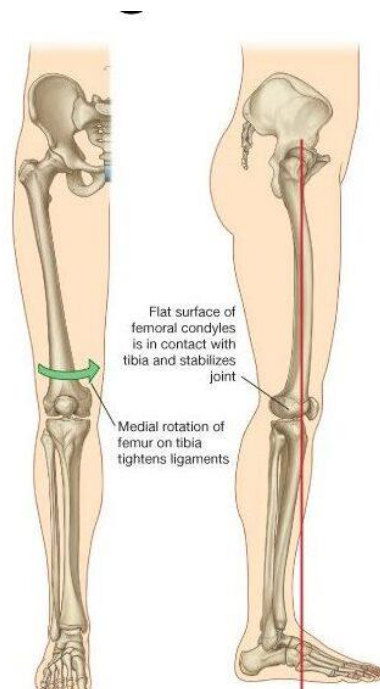
QUADRICEPS & POSTERIOR LIGAMENTS



- **Quadriceps femoris** stabilizes the knee anteriorly via the patella
- **Oblique popliteal ligament** (extension of semimembranosus):
 - Strengthens posterior capsule
 - Prevents hyperextension

LOCKING & UNLOCKING OF THE KNEE

LOCKING (STANDING UPRIGHT)



- Occurs at **terminal extension**
- Femur **medially rotates on tibia**
- Ligaments tighten
- Menisci compressed
- Knee becomes mechanically stable
- Allows standing with minimal muscle effort

This is why you don't collapse while standing.

UNLOCKING (INITIATING FLEXION)



- Done by **popliteus muscle**
- Popliteus:
 - Laterally rotates femur on tibia (or medially rotates tibia if non-weight bearing)
- Releases ligament tension
- Knee becomes mobile again

Popliteus is the **key that turns the lock**.

UNHAPPY (TERRIBLE) TRIAD

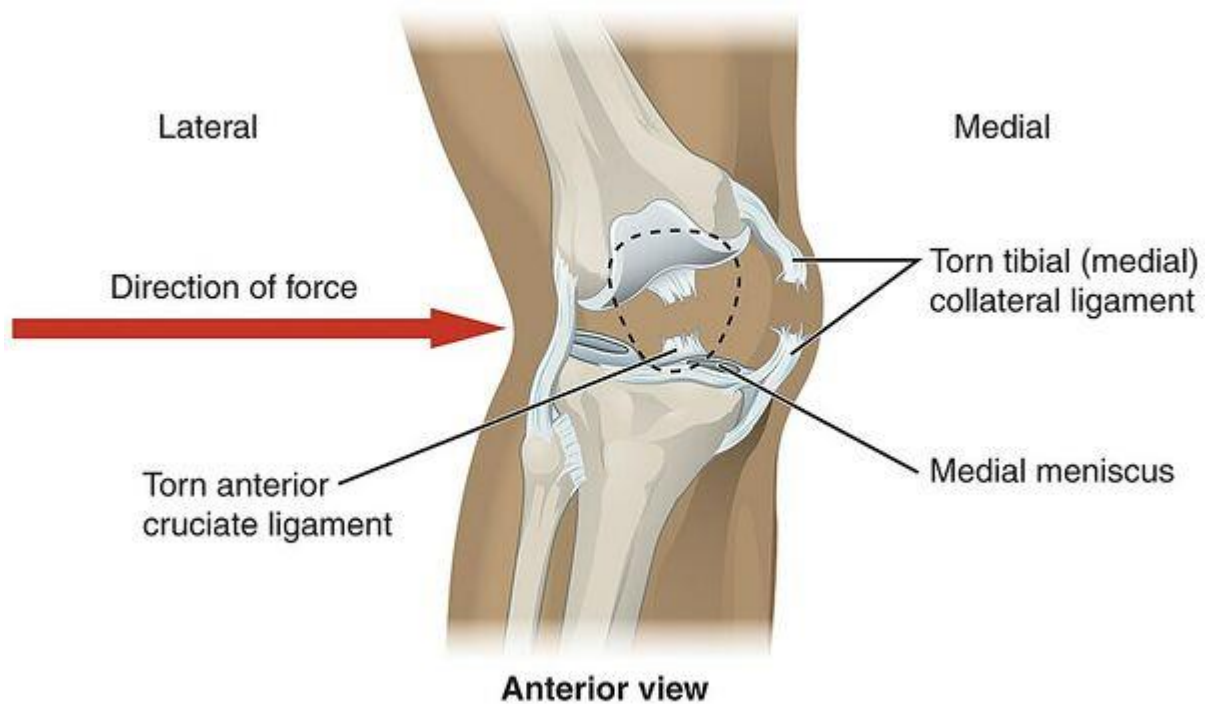
Classically involves:

- **ACL**
- **Medial Collateral Ligament**
- **Medial Meniscus**

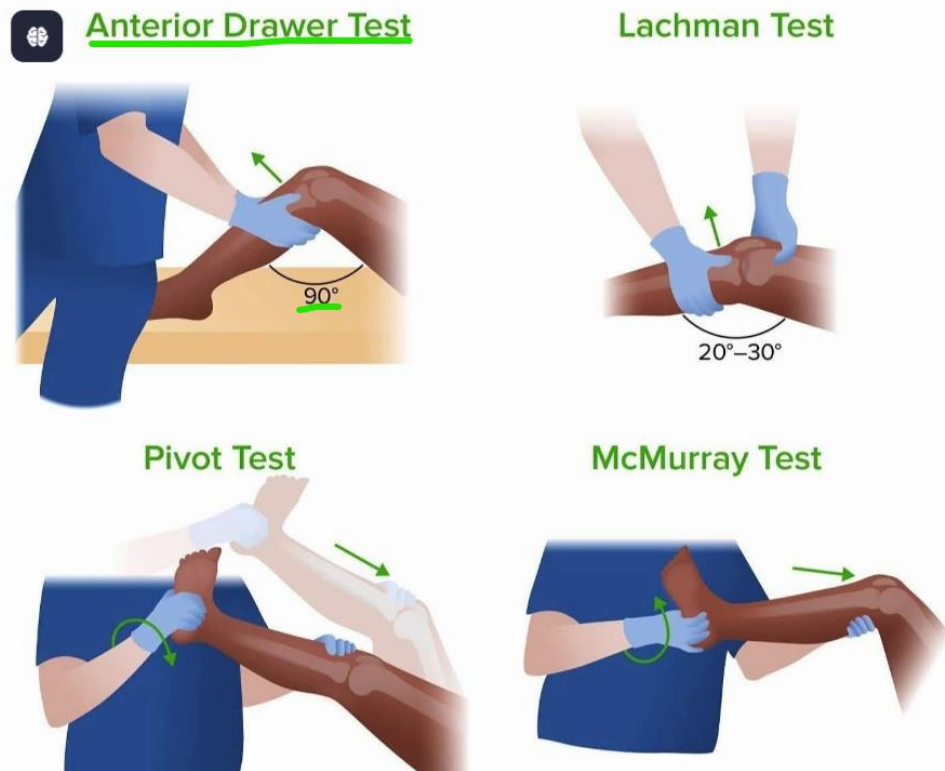
Mechanism:

- Lateral force to knee
- No involvement of fibula or LCL

This happens because the **medial side is structurally tethered**.



DRAWER TESTS (CRUCIATE LIGAMENT TESTING)



Patient lies supine, knee flexed to 90°, foot fixed.

ANTERIOR DRAWER TEST

- Pull tibia **anteriorly**
- Excessive forward movement → **ACL tear**

POSTERIOR DRAWER TEST

- Push tibia **posteriorly**
- Excessive backward movement → **PCL tear**

Key idea:

You are testing **tibia moving on femur**, not the other way around.

POPLITEAL FOSSA



The **popliteal fossa** is a **fat-filled, diamond-shaped space** behind the knee joint. It is not just an empty gap—it is a **protected passageway** that allows major vessels and nerves to change direction as the limb moves from thigh to leg.

Its purpose is threefold:

- Cushion neurovascular structures
- Allow knee flexion without compression
- Provide collateral circulation routes around the knee

This is why it's full of fat: fat is nature's shock absorber.

BOUNDARIES (WHY THE DIAMOND MAKES SENSE)

The fossa is shaped by muscles that **pull away during knee flexion**, opening space.

SUPERIOR BOUNDARIES (THIGH MUSCLES)

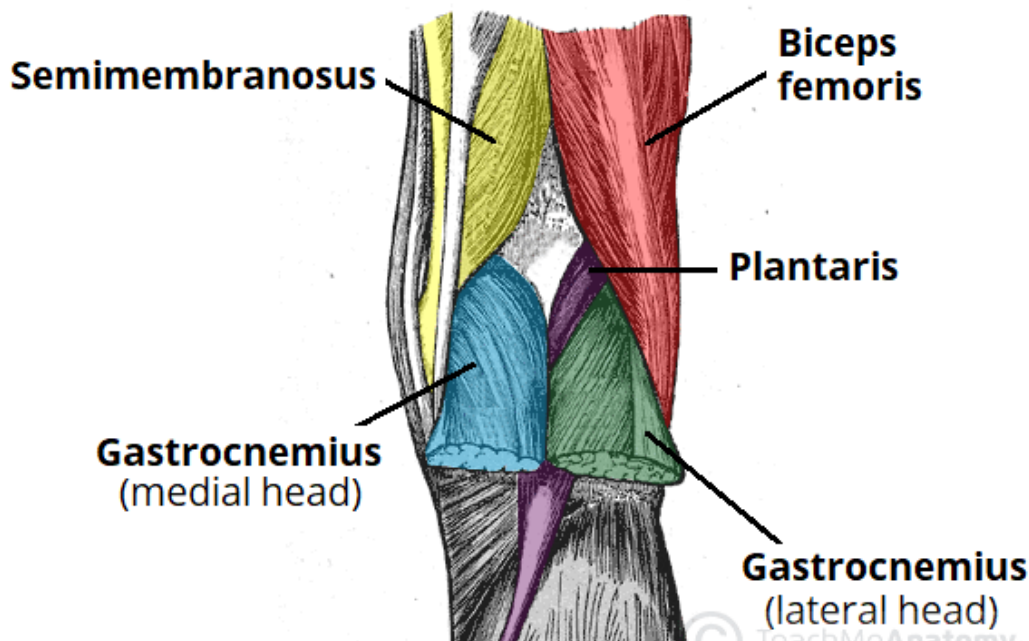
- **Superolateral: Biceps femoris**
- **Superomedial: Semimembranosus** (with semitendinosus superficial to it)

These hamstrings define the top of the diamond.

INFERIOR BOUNDARIES (LEG MUSCLES)

- **Inferolateral: Lateral head of gastrocnemius ± plantaris**
- **Inferomedial: Medial head of gastrocnemius**

Gastrocnemius forms the bottom of the diamond and explains why the fossa disappears when the calf contracts.



FLOOR

From superior → inferior, the floor is:

- **Popliteal surface of femur**
- **Posterior capsule of knee joint**
- **Oblique popliteal ligament**
- **Popliteus muscle**

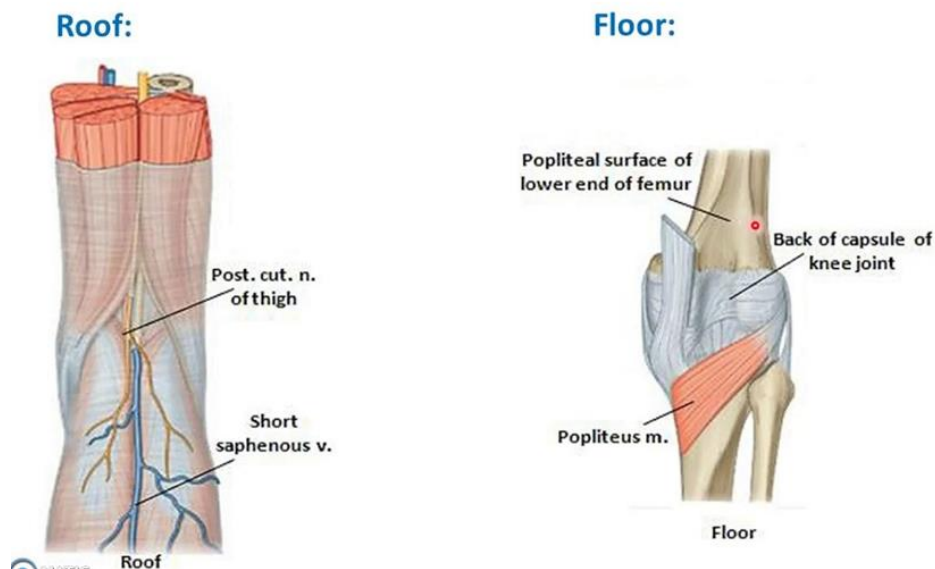
This matters clinically: knee joint infections can spread into the fossa.

ROOF

The roof is:

- Skin
- Superficial fascia
- Deep fascia (popliteal fascia)

This is why swelling here is visible early.



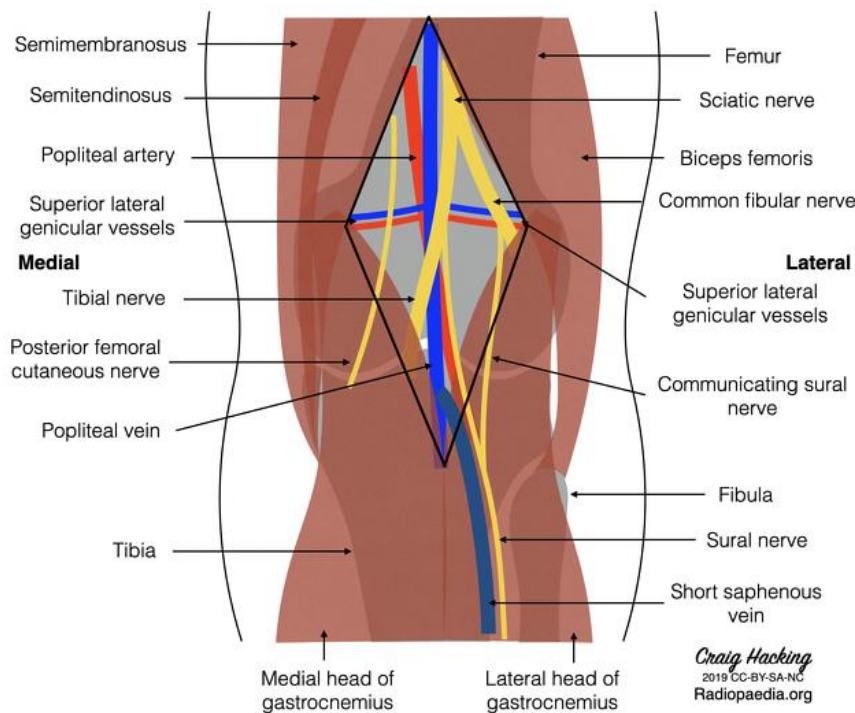
CONTENTS (AND WHY THEIR ORDER MATTERS)

The popliteal fossa contains **everything** that must survive **knee bending**.

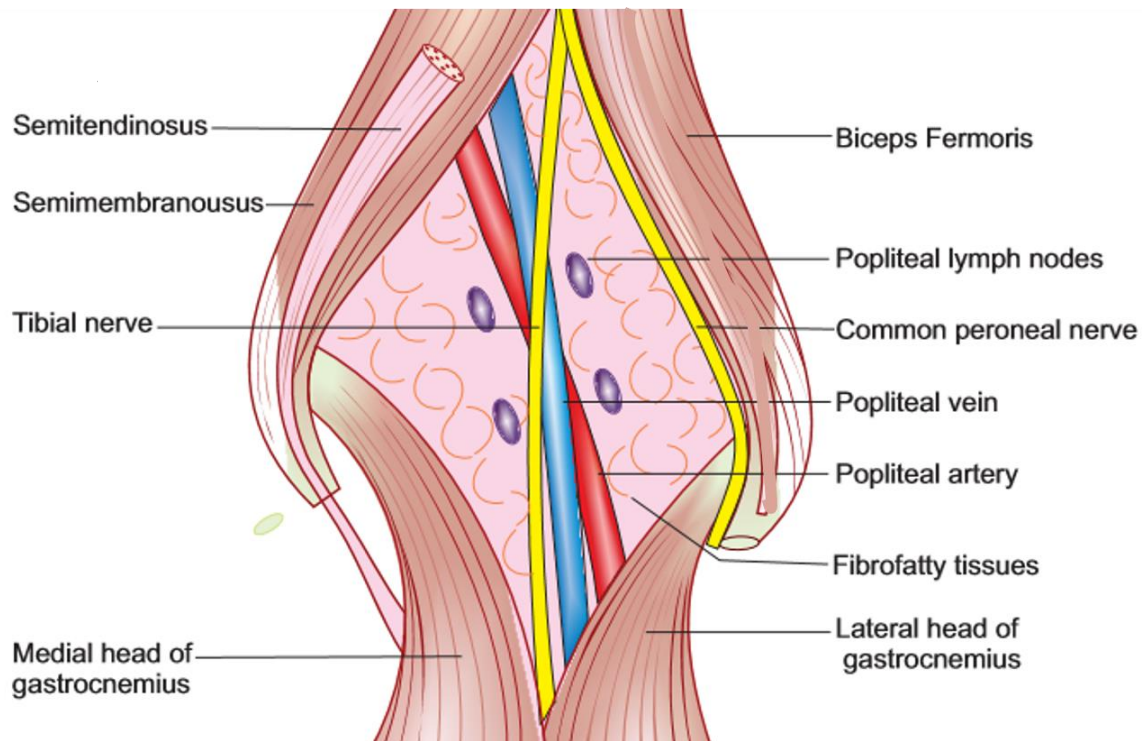
NEUROVASCULAR CONTENTS

- **Tibial nerve**
- **Common fibular (peroneal) nerve**
- **Popliteal vein**
- **Popliteal artery**
- **Popliteal lymph nodes**
- **Termination of small(Short) saphenous vein → popliteal vein**
- **Sural nerve (cutaneous)**
- **Superior Genicular Vessels**

POPLITEAL FOSSA



THE GOLDEN RULE: ARRANGEMENT FROM POSTERIOR TO ANTERIOR



This is one of the highest-yield facts in lower limb anatomy.

From **posterior (superficial)** → **anterior (deep)**:

N → V → A

- **N**erve is most superficial
- **V**ein is intermediate
- **A**rtery is deepest

CLINICAL INSIGHT

- The **popliteal artery is deepest**, so it is protected from trauma
- Palpating the popliteal pulse is difficult because the artery is pressed against bone and capsule
- A popliteal aneurysm compresses the vein and nerve first → swelling and neuropathy before ischemia

WHY THE POPLITEAL ARTERY BEHAVES DIFFERENTLY

The **femoral artery becomes the popliteal artery** after passing through the **adductor hiatus**.

In the fossa:

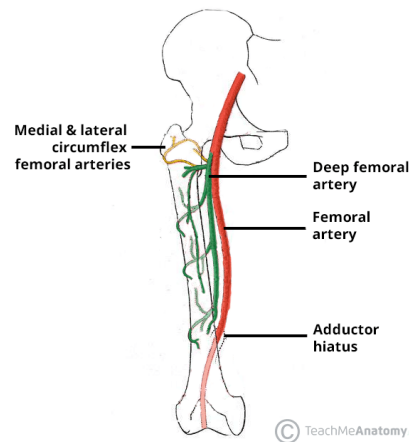
- It is tightly bound
- It cannot move freely
- It is vulnerable to knee dislocations

This is why knee dislocation is a **vascular emergency**.

ARTERIAL SUPPLY & COLLATERAL CIRCULATION (THE KNEE'S INSURANCE POLICY)

The lower limb depends mainly on the **femoral artery**. If it is blocked gradually, the limb often survives because of **collateral circulation**, especially around the knee.

DEEP FEMORAL ARTERY (PROFUNDA FEMORIS)

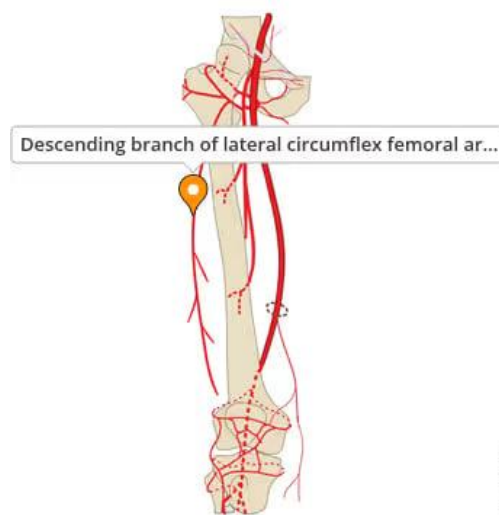


This is the **lifeline** during femoral artery blockage.

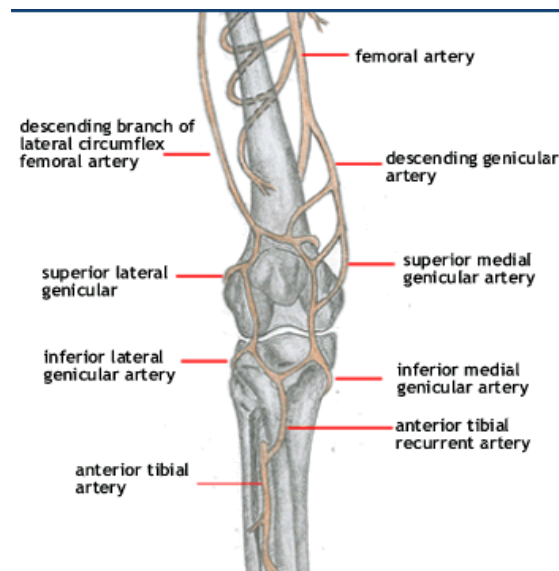
It gives:

- **Medial circumflex femoral artery**
- **Lateral circumflex femoral artery (LCFA)**
 - Ascending branch
 - Transverse branch
 - **Descending branch**

The **descending branch of LCFA** participates in **genicular anastomosis**.



GENICULAR ANASTOMOSIS (WHY THE KNEE DOESN'T DIE)



Around the knee:

Flexion for long time of period leads to artery kinking this will prevent perfusion of leg. Then the nature created arterial anastomosis to bypass this.

- Femoral artery branches (descending genicular, LCFA)
 - Popliteal artery branches (superior and inferior genicular arteries)
 - Anterior tibial recurrent artery
- Together, they form a **vascular ring**.

So if the femoral artery is obstructed at the **inguinal ligament**, blood can still reach the leg via:

- Profunda femoris
 - LCFA descending branch
 - Genicular arteries
 - Popliteal artery distal to blockage
- This is why **chronic** occlusion is survivable, but **acute** occlusion is catastrophic.

COMPARTMENTS OF THE LEG (LOWER LEG)

The leg is divided by deep fascia and intermuscular septa into **three compartments**.

1. ANTERIOR COMPARTMENT OF THE LEG

The **anterior compartment of the leg** is a **tight, osteofascial box** built for precision movements, not brute force.

LOCATION & BOUNDARIES (WHY COMPRESSION MATTERS)

It is located:

- **Anterior to:**
 - Intermuscular septum
 - Interosseous membrane
- **Between:**
 - Lateral surface of the shaft of **tibia**
 - Medial surface of the shaft of **fibula**
- **Bounded anteriorly by:**
 - Deep fascia of leg (crural fascia)
 - Skin

This deep fascia is **unyielding**, which explains why:

- Swelling here → **anterior compartment syndrome**
- Nerve and artery are easily compromised

FUNCTIONAL THEME OF THE ANTERIOR COMPARTMENT

Every muscle here does one main job:

☞ **Lift the foot and toes off the ground**

That means:

- **Dorsiflexion at ankle**
- **Extension of toes**

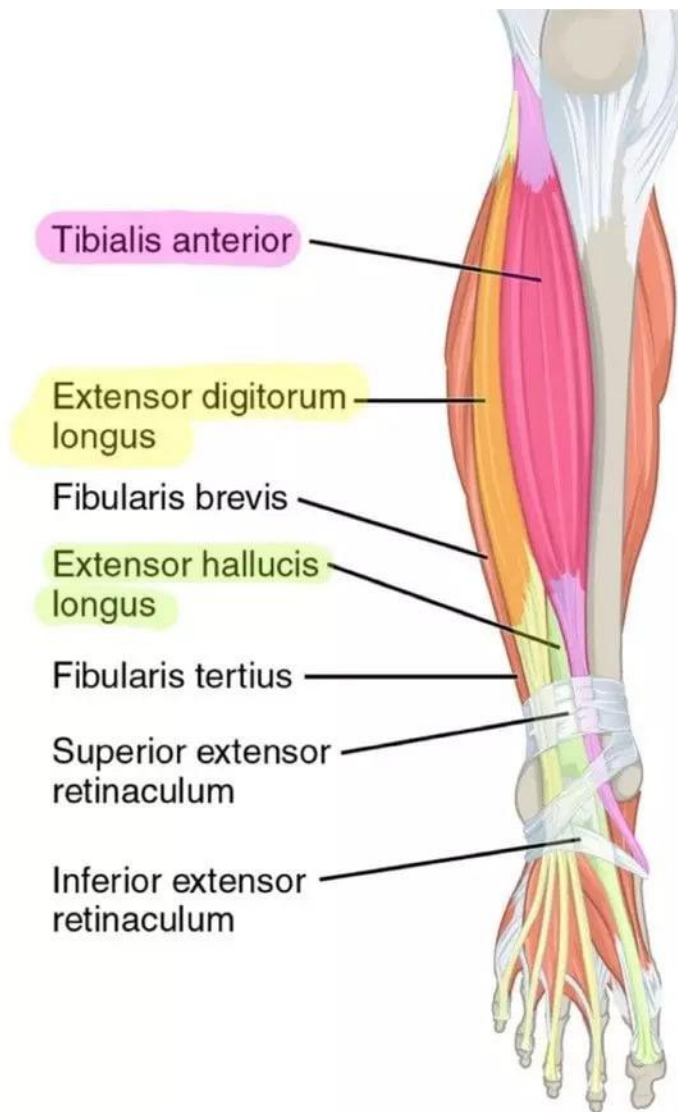
Why is this essential?

Because without it, your toes would drag during walking → **foot drop**.

3. CONTENTS OF THE ANTERIOR COMPARTMENT

MUSCLES (FROM MEDIAL → LATERAL)

1. **Tibialis anterior (TA)**
2. **Extensor hallucis longus (EHL) – deep**
3. **Extensor digitorum longus (EDL)**
4. **Fibularis (Peroneus) tertius – variable**



NEUROVASCULAR

- **Nerve:** Deep fibular (peroneal) nerve
- **Artery:** Anterior tibial artery

This pairing (deep fibular nerve + anterior tibial artery) is **constant and high-yield**.

1. TIBIALIS ANTERIOR — THE MASTER STABILIZER



POSITION

- **Most medial**
- **Most superficial**
- Forms the visible ridge beside the tibia

TENDON BEHAVIOR (IMPORTANT)

- Tendon appears halfway down the leg
- Passes:
 - Under **superior extensor retinaculum**
 - Under **inferior extensor retinaculum**
- Has its **own synovial sheath** → smooth movement

ATTACHMENTS

- **Origin:**
 - Lateral condyle of tibia
 - Superior half of lateral surface of tibia
 - Interosseous membrane
- **Insertion:**
 - Medial & inferior surfaces of **medial cuneiform**
 - Base of **1st metatarsal**

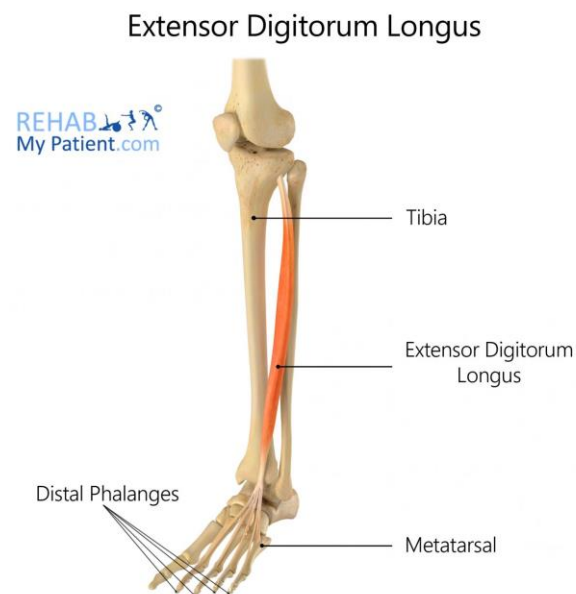
ACTIONS

- **Dorsiflexes** ankle
- **Inverts** foot (subtalar joint)
- **Maintains medial longitudinal arch** during walking

Clinical insight:

Loss of TA → flatfoot tendency + foot drop.

2. EXTENSOR DIGITORUM LONGUS (EDL)



POSITION

- **Most lateral** muscle of anterior compartment
- Partly overlapped by tibialis anterior

ATTACHMENTS

- **Origin:**
 - Lateral condyle of tibia
 - Superior $\frac{3}{4}$ of medial surface of fibula
 - Interosseous membrane
- **Insertion:**
 - Middle & distal phalanges of **lateral four toes**

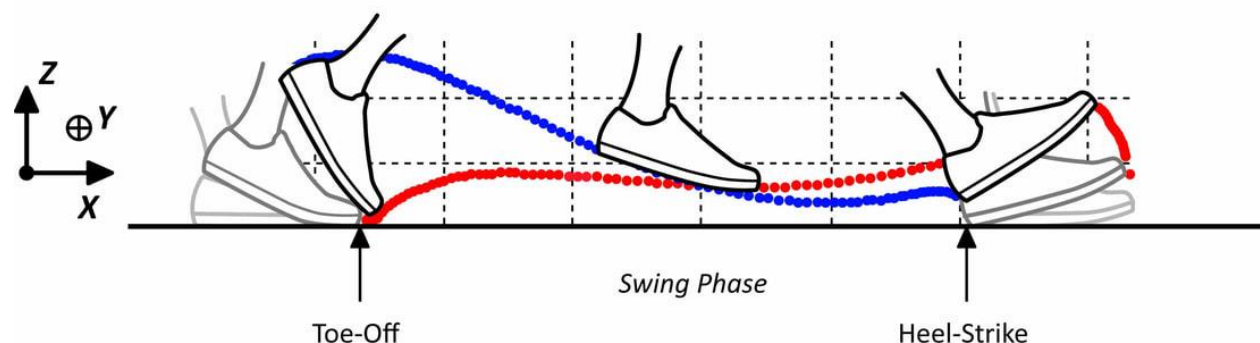
ACTION

- **Extends lateral four digits**
- **Dorsiflexes ankle**

Walking logic:

EDL clears the toes during **swing phase walking.**

Heel and Toe Clearance Pattern during gait



3. EXTENSOR HALLUCIS LONGUS (EHL) — HALLUCIS THE SAME NAME AS POLLICIS



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POSITION

- **Thin**
- **Deep**
- Lies between TA (medial) and EDL (lateral)

ATTACHMENTS

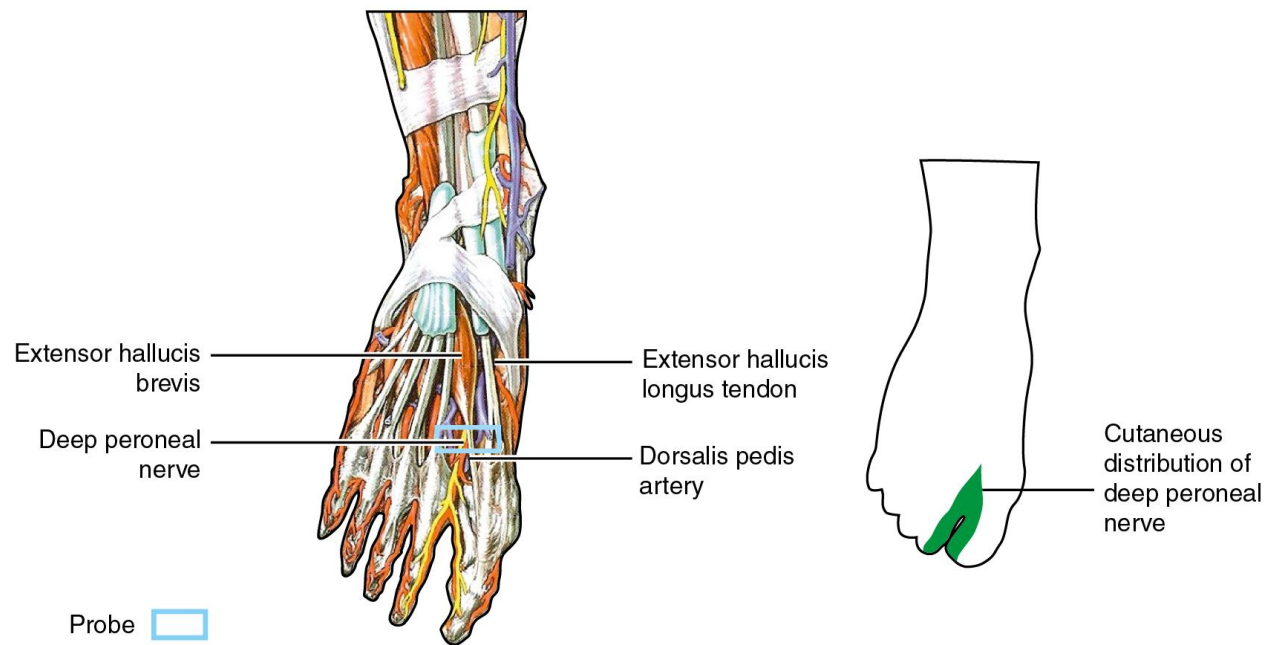
- **Origin:**
 - Middle part of anterior surface of fibula
 - Interosseous membrane
- **Insertion:**
 - Base of distal phalanx of great toe (dorsal surface)

ACTION

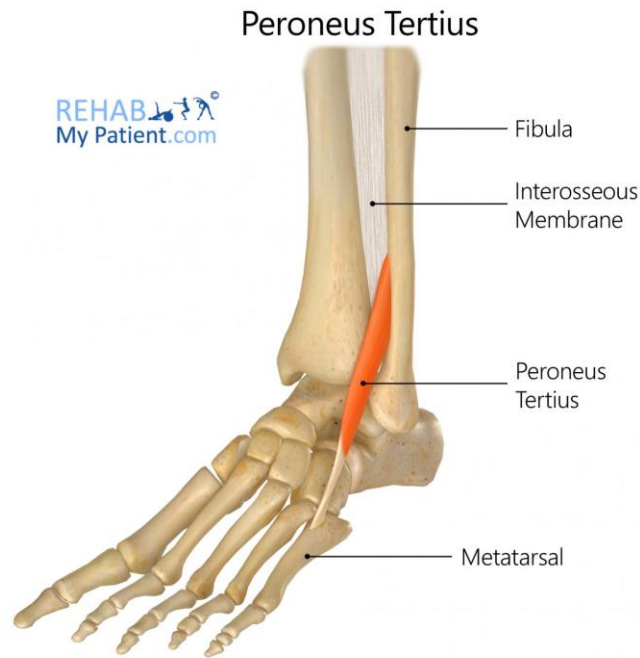
- Extends great toe
- Dorsiflexes ankle

Clinical pearl:

EHL tendon is used to test **deep fibular nerve function.**



4. FIBULARIS (PERONEUS) TERTIUS — THE OPTIONAL MUSCLE



NATURE

- A **separated part of EDL**
- **Not always present** (exam favorite)

ATTACHMENTS

- **Origin:**
 - Inferior third of anterior surface of fibula
 - Interosseous membrane
- **Insertion:**
 - Dorsum of base of **5th metatarsal**

ACTION

- **Dorsiflexes ankle**
- **Everts foot**

It acts as a **counterbalance**, preventing excessive inversion during dorsiflexion.

NERVE SUPPLY — DEEP FIBULAR (PERONEAL) NERVE

ROOTS

- **L4, L5**

ORIGIN

- Branch of **common fibular nerve**

COURSE (HIGH-YIELD PATHWAY)

- Enters anterior compartment
- Descends **deep to EDL**
- Relation to artery:
 - Initially lateral
 - Then anterior
 - Finally **lateral to anterior tibial artery**
- Crosses distal tibia
- Enters dorsum of foot

SUPPLIES

- **All anterior compartment muscles**
- **Intrinsic foot extensors (EDB, EHB)**
- **Skin:** only **first interdigital cleft**

- **Articular branches:** ankle joint

Single sensory patch = huge exam clue.

COMMON FIBULAR (PERONEAL) NERVE — THE VULNERABLE ONE

ROOTS

- **L4–S2**
- Posterior divisions

COURSE

- Runs along medial border of **biceps femoris**
- Leaves popliteal fossa laterally
- Winds around **neck of fibula** (very superficial → injury risk)
- Pierces **peroneus longus**
- Divides into:
 - **Superficial fibular nerve**
 - **Deep fibular nerve**

BRANCHES

- **Muscular:** short head of biceps femoris
- **Cutaneous:**
 - Lateral sural communicating nerve
 - Lateral cutaneous nerve of calf

Fracture neck of fibula → **foot drop**.

ARTERIAL SUPPLY — ANTERIOR TIBIAL ARTERY

ORIGIN

- Branch of **popliteal artery**

COURSE

- Enters anterior compartment via interosseous membrane
- Runs with **deep fibular nerve**
- Passes:
 - Behind superior extensor retinaculum
 - **Lateral to EHL tendon**
 - **Medial to deep fibular nerve & EDL tendons**
- Crosses ankle joint

TERMINATION

- Continues as **dorsalis pedis artery**

Pulse here = dorsalis pedis pulse.

SEE LATERAL AND POSTERIOR COMPARTMENT MUSCLES BY YOURSELF!

LATERAL COMPARTMENT

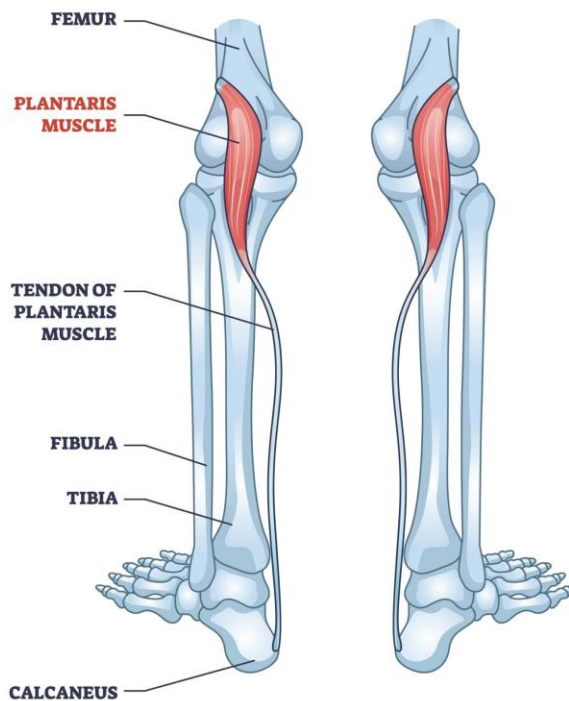
- Common action: Evertors
- **Nerve: Superficial fibular nerve**
- **Artery: Supplies most of the dorsum of foot**

POSTERIOR COMPARTMENT

- Common Action: Plantarflexors
- Nerve: Tibial nerve
- Artery: Posterior tibial artery

PLANTARIS MUSCLE (FRESHMAN'S NERVE)

PLANTARIS MUSCLE



Plantaris has:

- A tiny muscle belly
- An extremely long, thin tendon

This tendon **looks exactly like a nerve.**

Hence the nickname: **“Freshman’s nerve”**
Medical students cut it thinking it’s useless. Surgeons harvest it for grafts.

Functionally? Minimal.
Anatomically? Legendary.

ANKLE RETINACULA



The retinacula of the ankle are **localized thickenings of deep fascia**.

Their jobs are beautifully simple:

- **Hold long tendons close to bone**
- **Prevent bowstringing** during dorsiflexion
- **Act as pulleys**, improving efficiency of muscle force

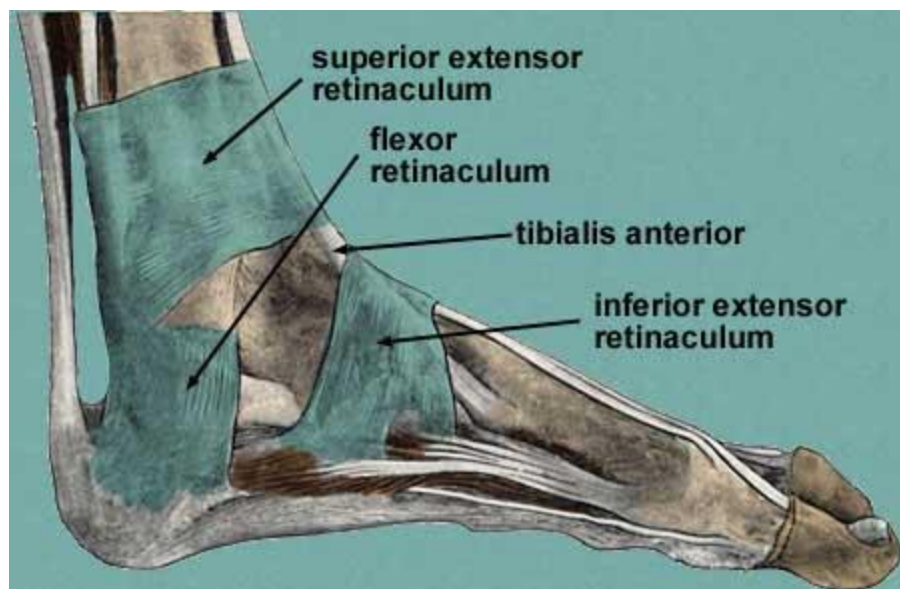
Without retinacula, dorsiflexion would be weak, sloppy, and painful.

EXTENSOR RETINACULA = ANTERIOR ANKLE STABILIZERS

There are **two extensor retinacula** on the anterior side of the ankle:

1. **Superior extensor retinaculum**
2. **Inferior extensor retinaculum**

They work together but are anatomically distinct.



SUPERIOR EXTENSOR RETINACULUM

SHAPE & POSITION

- **Horizontal band** of deep fascia
- Lies just **above the ankle joint**

ATTACHMENTS

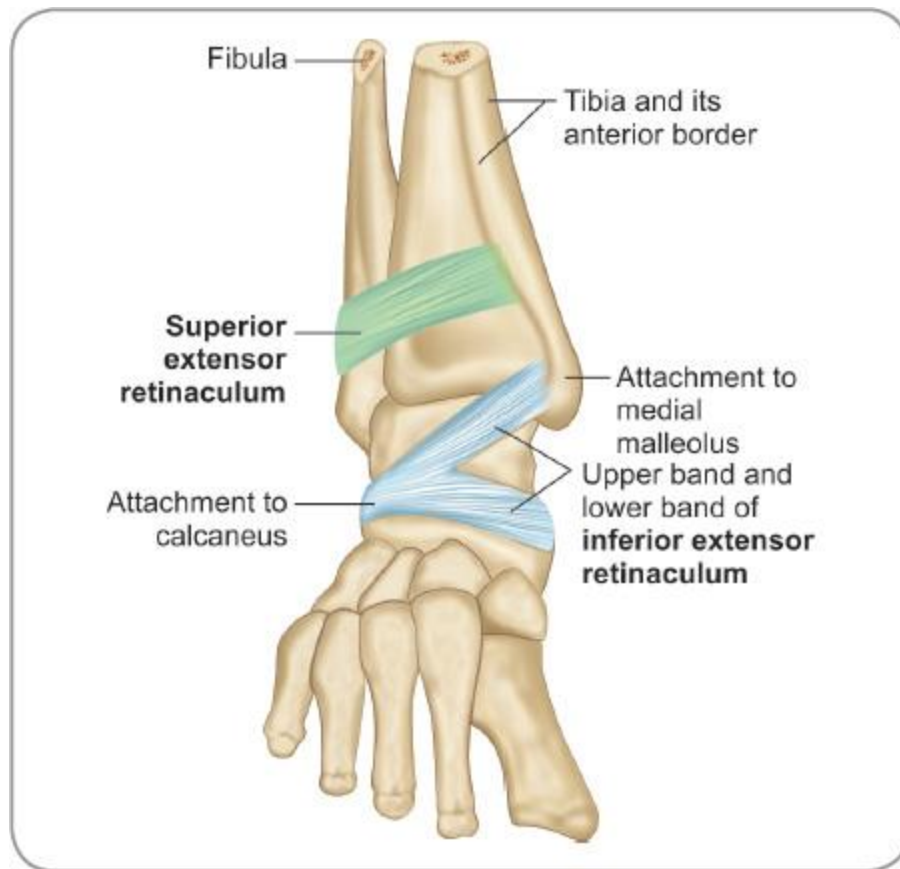
- Medially → **anterior border of tibia**
- Laterally → **anterior border of fibula**
- Both at their **distal ends**

Think of it as a **belt across the front of the ankle.**

FUNCTION

- Holds extensor tendons in position before they cross the ankle joint
- Prevents bowstringing during dorsiflexion

INFERIOR EXTENSOR RETINACULUM (THE IMPORTANT ONE)



SHAPE

- **Y-shaped**
- Much stronger and more complex

LATERAL ATTACHMENT

- **Anterosuperior surface of calcaneus**

This is the stem of the “Y”.

MEDIAL BANDS

- **Upper bands** → attaches to **medial malleolus**
- **Lower bands** → blends with **plantar fascia** along the medial border of foot

STRUCTURES PASSING in front of THE EXTENSOR RETINACULA

These are **superficial structures** and do **not** need retinacular restraint.

IN FRONT OF **MEDIAL MALLEOLUS**

- **Great saphenous vein**
- **Saphenous nerve**

ACROSS THE ANKLE DORSUM

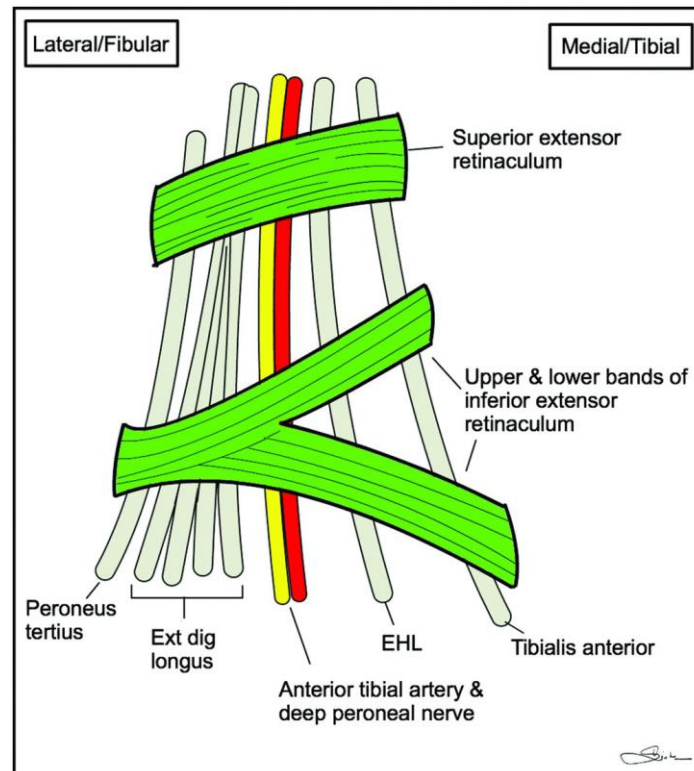
- **Superficial fibular (peroneal) nerve**
 - Medial branch
 - Lateral branch

These structures are subcutaneous, not load-bearing.

6. STRUCTURES PASSING **beneath(Inside)** THE EXTENSOR RETINACULA

(Extremely high-yield)

From **medial** → **lateral**, remember this sequence:



T E A D E P

Let's decode it properly.

1. **Tibialis anterior tendon**
2. **Extensor hallucis longus tendon**
3. **Anterior tibial artery (+ venae comitantes)**
4. **Deep fibular (peroneal) nerve**
5. **Extensor digitorum longus tendons**
6. **Fibularis (peroneus) tertius tendon**

This order is **constant** and clinically critical.

7. WHY THIS ORDER MATTERS

- The **artery and nerve** lie between the long toe extensors → protected but vulnerable to compression
- Tendon sheath inflammation (tenosynovitis) can compress the **deep fibular nerve**
- Surgical approaches to ankle avoid this bundle by respecting the order

Mnemonic logic:

- **TA** is most medial because it supports the medial arch
- **PT (fibularis tertius)** is lateral because it assists eversion

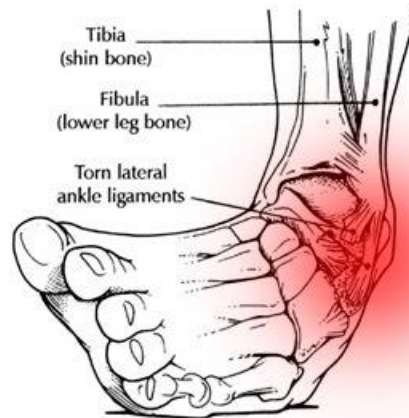
8. FUNCTIONAL SYNTHESIS

When you dorsiflex your foot:

- Extensor muscles pull
- Retinacula hold tendons down
- Force is transmitted efficiently
- No bowstringing
- No tendon snapping
- No nerve or vessel kinking(**we don't say vessel bending**)

The ankle retinacula turn **linear muscle pull** into **controlled rotation**.

OVER-INVERSION



- Most common
- Lateral ligaments injured
- Superficial fibular nerve affected
- Peroneal retinacula strained
- Dorsum of foot sensory loss

OVER-EVERSION

- Less common, more severe
- Deltoid ligament or medial malleolus involved
- Tibial nerve compression (tarsal tunnel)
- Flexor retinaculum implicated
- Sole of foot sensory symptoms



THE END